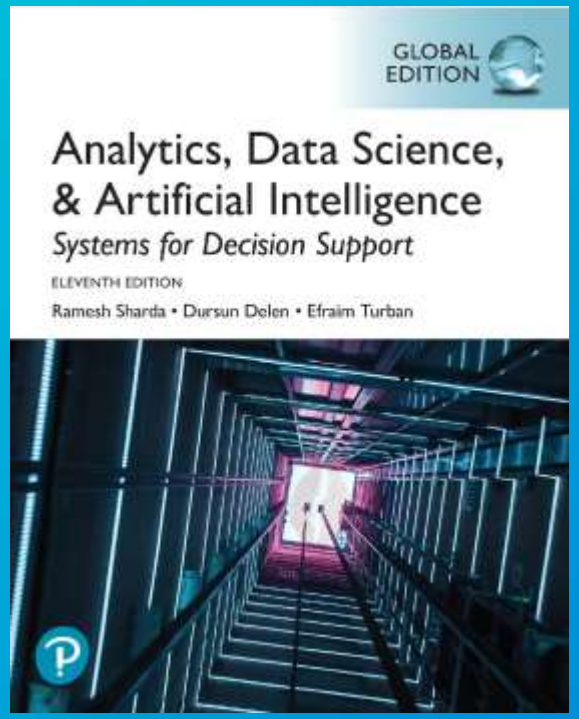
MANAGEMENT SUPPORT SYSTEMS
(415 MIS-3)
CHAPTER – 1



Chapter 1: Overview of Business Intelligence, Analytics, Data Science, and Artificial Intelligence: Systems for Decision Support

Analytics, Data Science, & Artificial Intelligence Systems For Decision Support,11ed, Global Edition

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- **1.3** - Decision-Making Processes and Computer Decision Support Framework
- **1.4** - Evolution of Computerized Decision Support to Business Intelligence/Analytics/Data Science
- **1.5** - Analytics Overview
- **1.6** - Analytics Examples in Selected Domains

1.2 Changing Business Environments and Evolving Needs for Decision Support and Analytics

- The Decision Making Process
- The Influence of the External and Internal Environments on the Process
- Data and Its Analysis in Decision Making
- Technologies for Data Analysis and Decision Support

The Decision Making Process

- The business world is full of uncertainties and rapid changes.
- Thus, the decision making process can determine the success or failure of an organization, and how well it performs.
- In the past decision making was based on creativity, instinct, and experience.
- Now, decision making is more grounded in scientific approach, and utilizes systematic quantitative methods.

The Decision Making Process (cont.)

Managers usually undergo the decision making process using the following steps:

1. Understand the decision you have to make
2. Collect all the information *(Define the problem)*
3. Identify the alternatives *(Construct a model & Identify possible solutions)*
4. Evaluate the pros and cons
5. Select the best alternative *(Compare, choose and recommend the best solution)*
6. Make the decision
7. Evaluate the impact of your decision

The Influence of External & Internal Environments

- Predicting the consequences and the future of any given decisions is complex. This is due to the uncertainty that can arise from multiple factors including:
    1. Political Factors (E.g. government policies, political instability)
    2. Economic Factors (E.g. competition, changing demand)
    3. Sociological and psychological factors
    4. Environment Factors

- Because of these constant changes, the trial-and-error approach to management is unreliable and unsustainable. Managers must begin to use the new tools and techniques of their fields.

Data and Its Analysis in Decision Making

- The amount of data doubles every two years.
- Decision making process requires an organization to collect and analyze vast stores of data.
- Computer applications have moved from

  transaction-processing and monitoring activities => to

  problem analysis and solution applications.
- These activities are done with cloud-based technologies, mostly accessed through mobile devices.

Data and Its Analysis in Decision Making (cont.)

- The foundation of modern management lies in

    1. analytics and BI tools such as data warehousing, data mining, online analytical processing (OLAP), dashboards, and

    2. the use of cloud-based systems for decision support.

Technologies for Data Analysis and Decision Support

The following developments have contributed to the growth of decision support and analytics technologies:

1. Group communication and collaboration
2. Improved data management
3. Managing giant data warehouses & Big Data
4. Analytical Support
5. Overcoming cognitive limits in processing and storing information
6. Knowledge Management
7. Anywhere, anytime support
8. Innovation and artificial intelligence

1.3 Decision-Making Processes and Computer Decision Support Framework

- Simon's Process: Intelligence, Design, Choice, and Implementation Phases
- The Classical Decision Support System Framework
- DSS Application
- Characteristics of DSS
- Components of a Decision Support System

Simon's Process: Intelligence, Design, and Choice

Decision-making process phases:

1. **Intelligence:** The decision maker examines reality, identifies and defines the problem.
2. **Design:** A model representing the system is constructed by ⬇ and then validated.
    1. making assumptions to simplify reality and
    2. identifying relationships between variables.
3. **Choice:** Selection of a proposed solution to the model . This solution is tested to determine its viability.
4. **Implementation:** Successful implementation results in solving the real problem. Failure leads to a return to an earlier phase of the process.

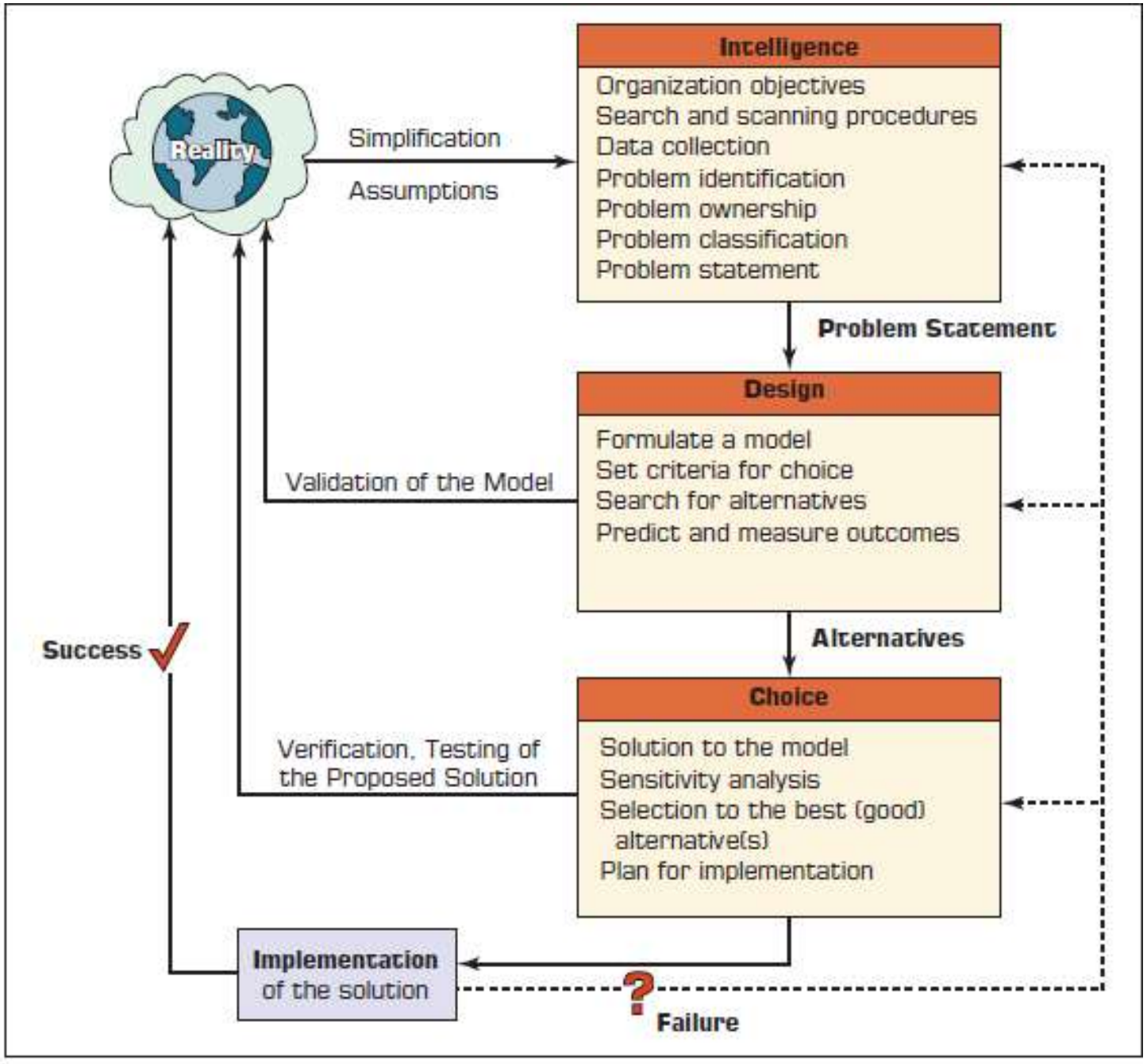


FIGURE 1.1 The Decision-Making/Modeling Process.

The Classical Decision Support System Framework

The **framework** for computerized decision support can be represented in a 3x3 **matrix** with two dimensions:

1. **Degrees of structure**:
   - **Structured:** Routine/repetitive problems for which standard solution methods exist
   - **Semi structured:** Fall between structured and unstructured problems
   - **Unstructured:** Complex problems where there are no clear, standard solution methods
2. **Types of controls:**
   - **Strategic planning:** Defining long-range goals/policies for resource allocation
   - **Management control:** The efficient use of resources in the accomplishment of organizational goals
   - **Operational control:** the efficient and effective execution of specific tasks

Type of Decision	Type of Control: Operational Control	Managerial Control	Strategic Planning
Structured	1 Monitoring accounts receivable Monitoring accounts payable Placing order entries	2 Analyzing budget Forecasting short-term Reporting on personnel Making or buying	3 Managing finances Monitoring investment portfolio Locating warehouse Monitoring distribution systems
Semistructured	4 Scheduling production Controlling inventory	5 Evaluating credit Preparing budget Laying out plant Scheduling project Designing reward system Categorizing inventory	6 Building a new plant Planning mergers and acquisitions Planning new products Planning compensation Providing quality assurance Establishing human resources policies Planning inventory
Unstructured	7 Buying software Approving loans Operating a help desk Selecting a cover for a magazine	8 Negotiating Recruiting an executive Buying hardware Lobbying	9 Planning research and development Developing new technologies Planning social responsibility

FIGURE 1.2 Decision Support Frameworks.

DSS Application

Key difference between DSS and BI applications:

1. **Business intelligence (BI) systems** monitor situations and identify problems and/or opportunities using analytic methods.
2. **DSS** is a methodology for supporting decision making:-
    - It uses an interactive, adaptable **computer-based information system** (CBIS) developed for supporting the solution to a specific unstructured management problem.
    - It uses **data**, provides a user-friendly **interface**, and incorporates the **decision maker's insights**.
    - It includes **models** that developed through an interactive and iterative process.

Characteristics of DSS

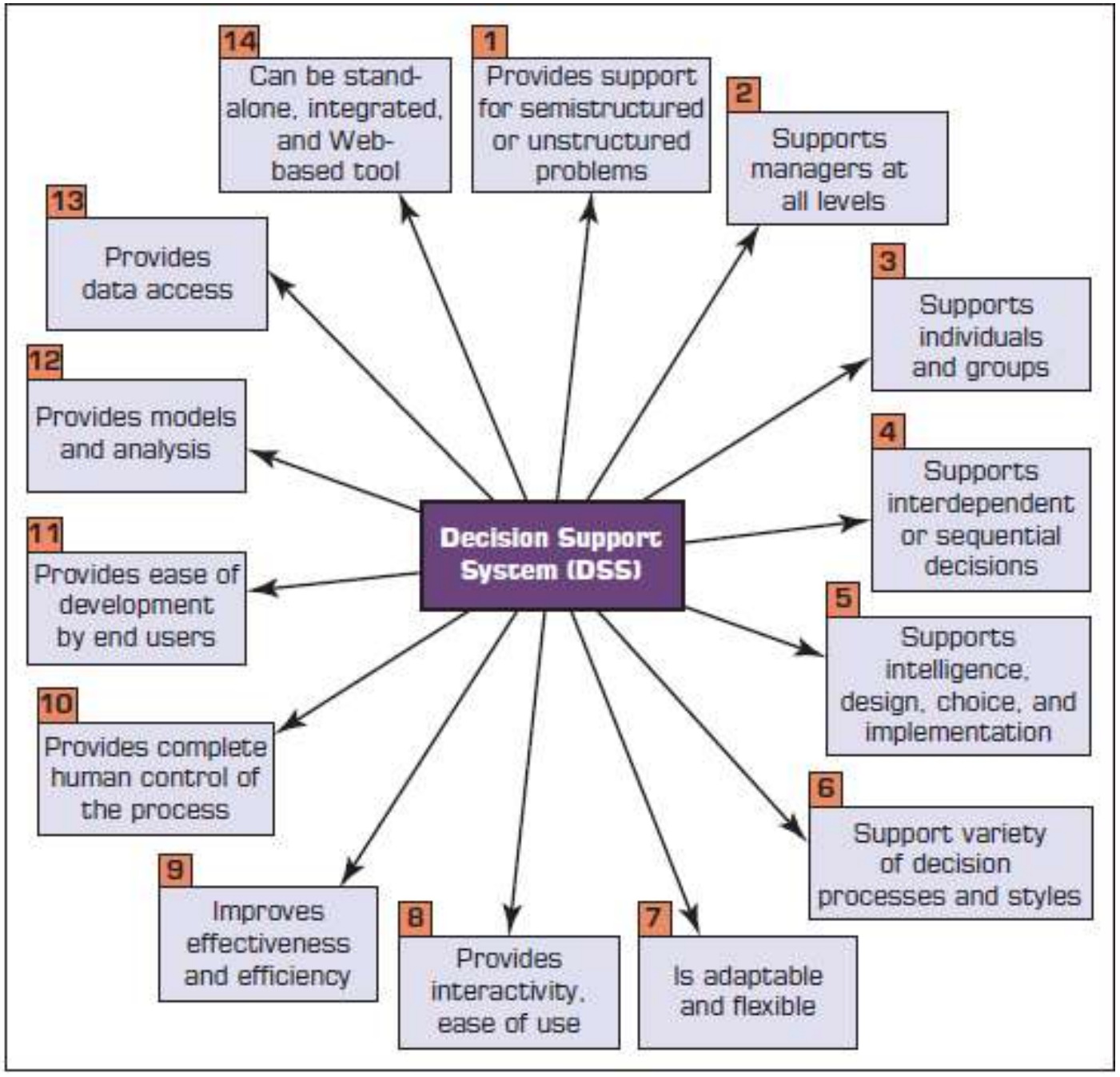


FIGURE 1.3 Key Characteristics and Capabilities of DSS.

Components of a Decision Support System

A DSS can be composed of:

1. Data management subsystem
2. Model management subsystem
3. User interface subsystem
4. Knowledge-based management subsystem

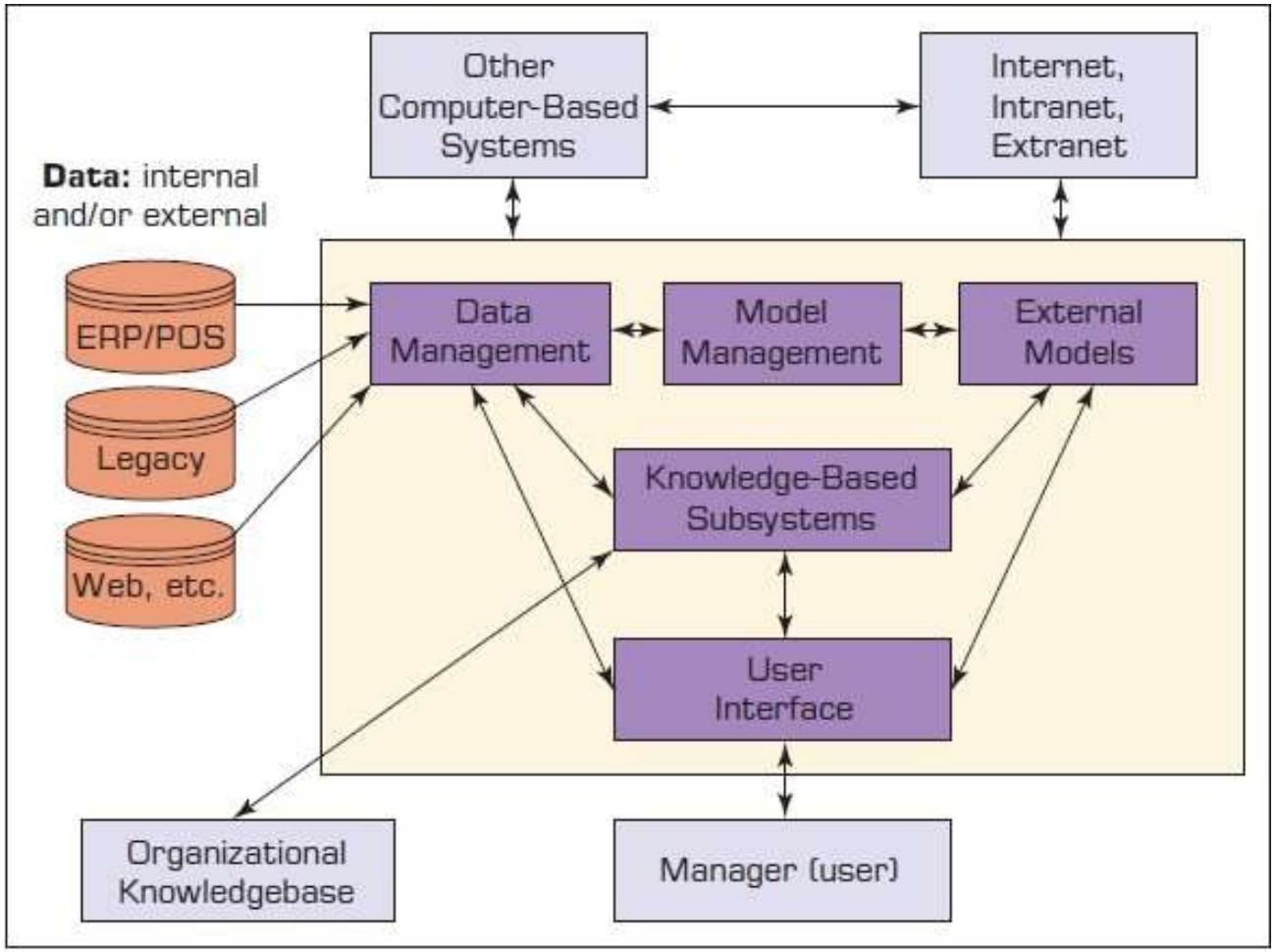


FIGURE 1.4 Schematic View of DSS.

1. Data Management Subsystem

The data management subsystem is composed of the following elements:

1. DSS database
2. Database management system
3. Data directory
4. Query facility

- It can be interconnected with the corporate data warehouse (a repository for corporate relevant decision-making data)

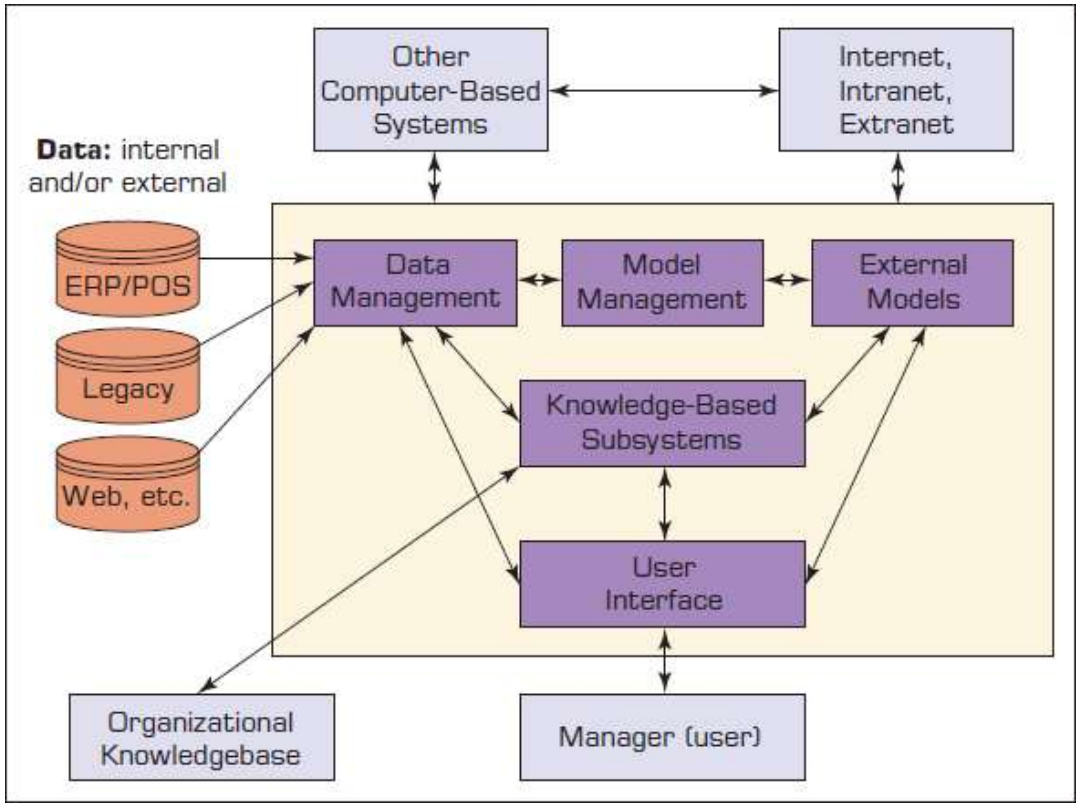


FIGURE 1.4 Schematic View of DSS.

2. Model Management Subsystem

The model management subsystem is a component that includes quantitative models that provide the

- system's analytical capabilities and
- appropriate software management.

The model management subsystem of a DSS is composed of the following elements:

1. Model base
2. MBMS (Model Base Management System)
3. Modeling language
4. Model directory
5. Model execution, integration, and command processor

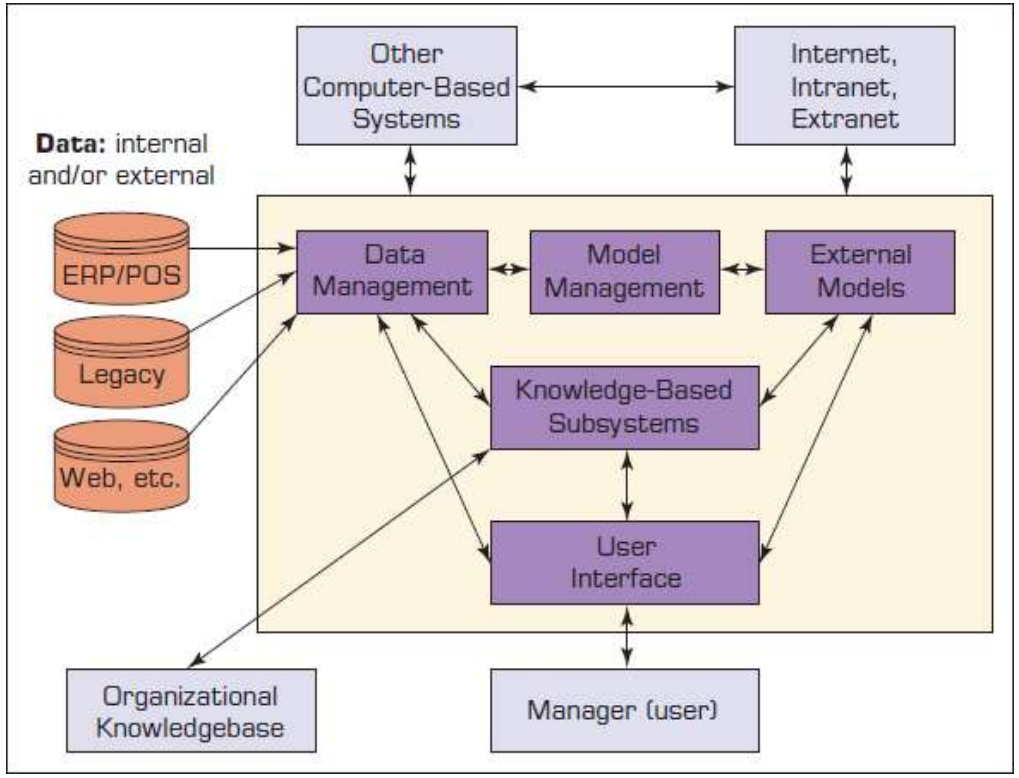


FIGURE 1.4 Schematic View of DSS.

3. The User Interface Subsystem

- The user is considered part of the system.
- The user communicates with and commands the DSS through the user interface subsystem.
- The Web browser has been recognized as an effective DSS GUI because it is
  - flexible,
  - user-friendly, and a
  - gateway to almost all sources of necessary information and data.
- Web browsers have led to the development of portals and dashboards (front end of many DSS).

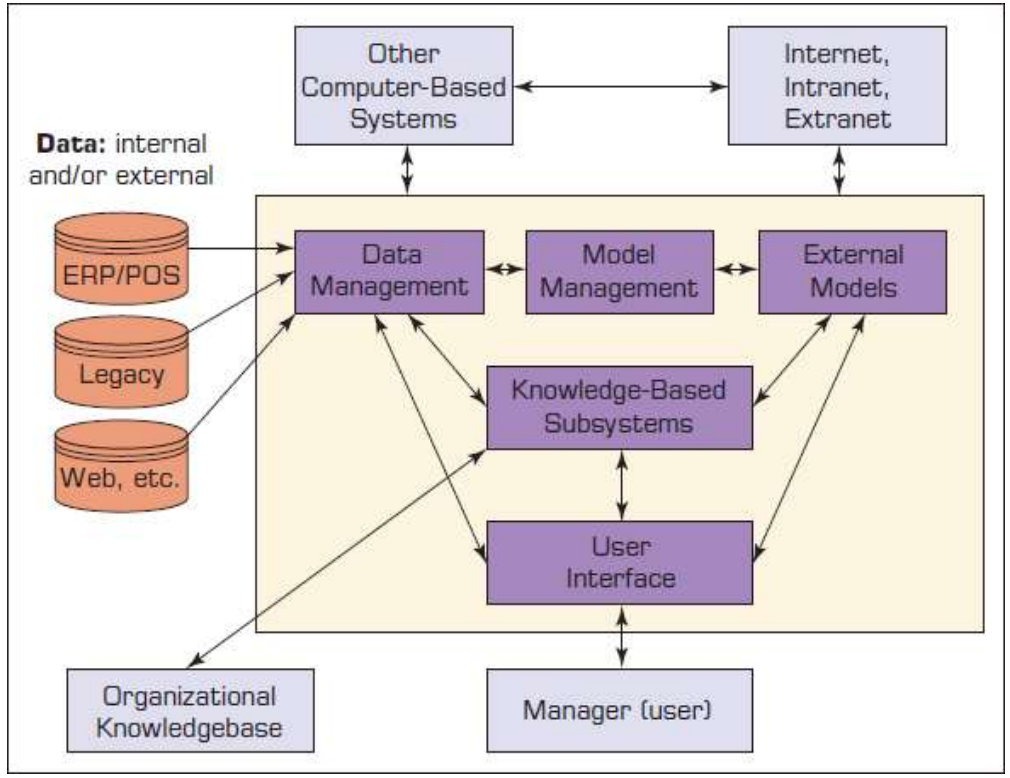


FIGURE 1.4 Schematic View of DSS.

4. The Knowledge-Based Management Subsystem

- The knowledge-based management subsystem provides intelligence
  - to augment the decision maker's own intelligence
  - to help understand a user's query and provide a consistent answer.

- Interconnected with the organization's knowledge repository or connected to thousands of external knowledge sources.

- User interface developments are closely tied to the major new advances in their knowledge-based systems.

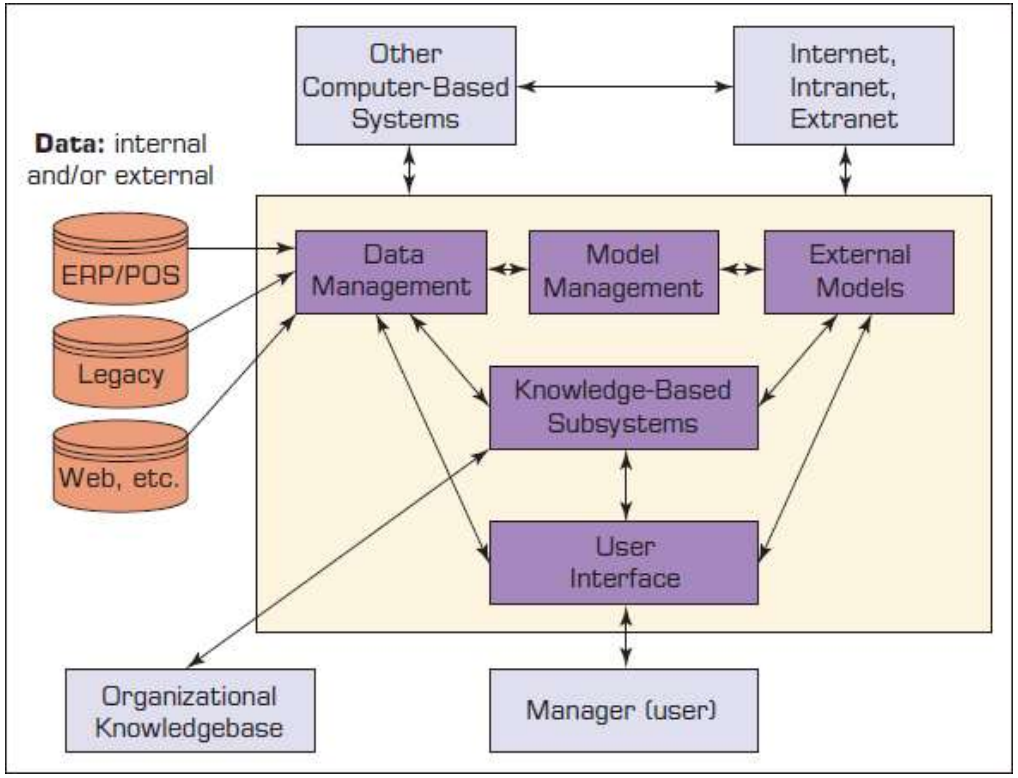


FIGURE 1.4 Schematic View of DSS.

1.4 Evolution of Computerized Decision Support to Business Intelligence/Analytics/Data Science

- Evolution Of Computerized Decision Support
- Framework for Business Intelligence
- Architecture of BI
- The Origins and Drivers of BI
- Data Warehouse as a Foundation for Business Intelligence
- Transaction Processing versus Analytic Processing
- A Multimedia Exercise in Business Intelligence

Evolution Of Computerized Decision Support

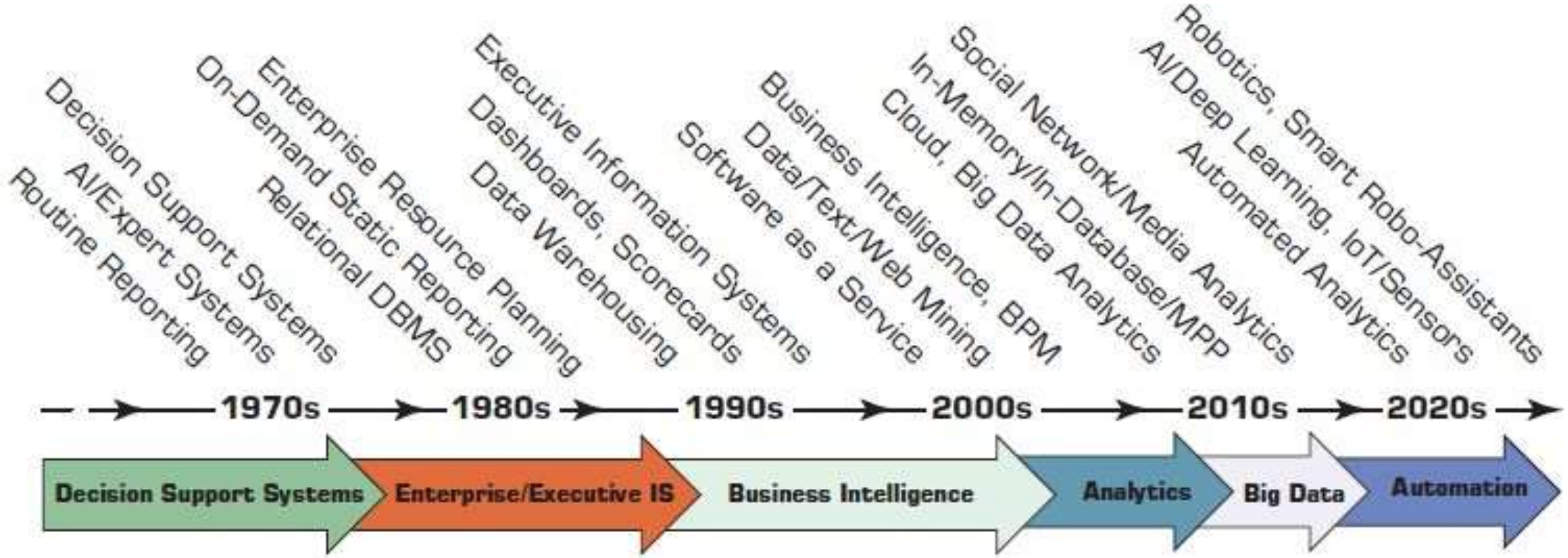


FIGURE 1.5 Evolution of Decision Support, Business Intelligence, Analytics, and AI.

Evolution Of Computerized Decision Support (cont.)

1. **Management Information Systems (MIS):** a variety of reports used to understand and address the changing business needs and challenges.

2. **Decision Support Systems (DSS):** combination of individuals intellectual resources with the capabilities of the computer to improve the quality of decisions.
   - DSS designed and developed specifically for executives and their decision-making needs

3. **Executive Information Systems (EIS):** the need for more versatile reporting leads to the development of EISs;
   - These systems were designed as graphical dashboards that allow decision makers to keep track of the key performance indicators.

A Framework for Business Intelligence (BI)

- **BI** is an umbrella term that combines architectures, tools, databases, analytical tools, applications, and methodologies.
- **BI process** is based on the transformation of data -> information -> decisions -> actions
- BI's major **objective** is to enable
  1. interactive access to data (in real time),
  2. data manipulation,
  3. appropriate analyses.
- By analyzing **historical** and **current** data, **situations**, and **performances**, decision makers able to make more informed and better decisions.

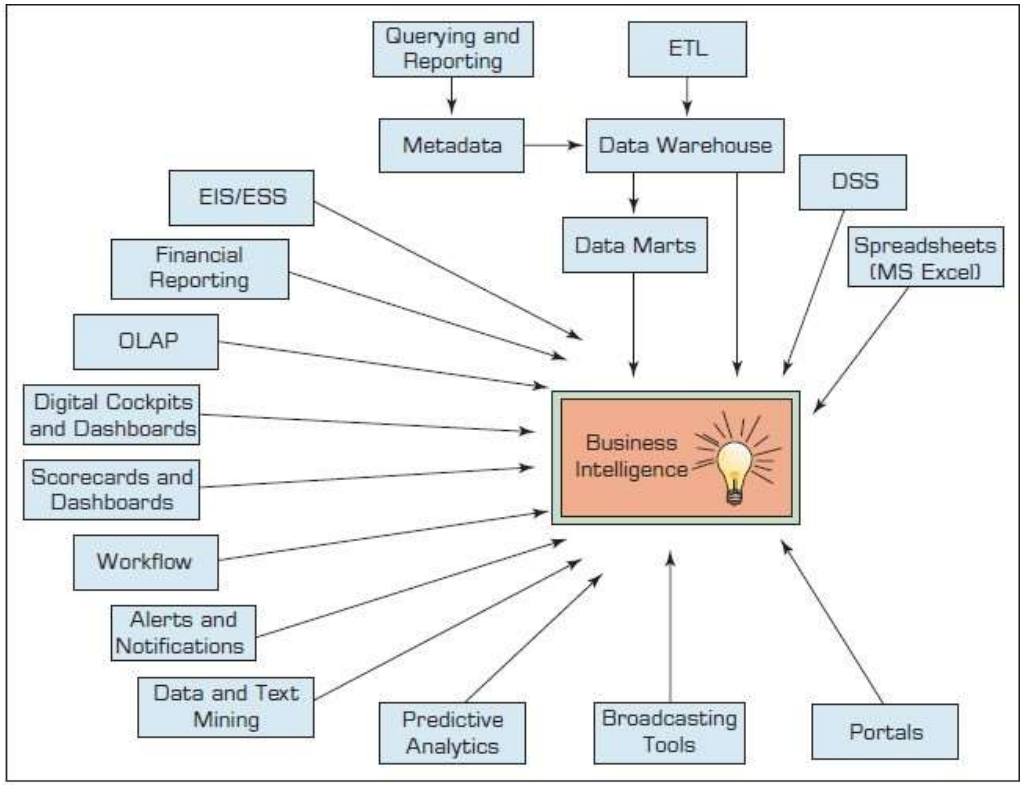
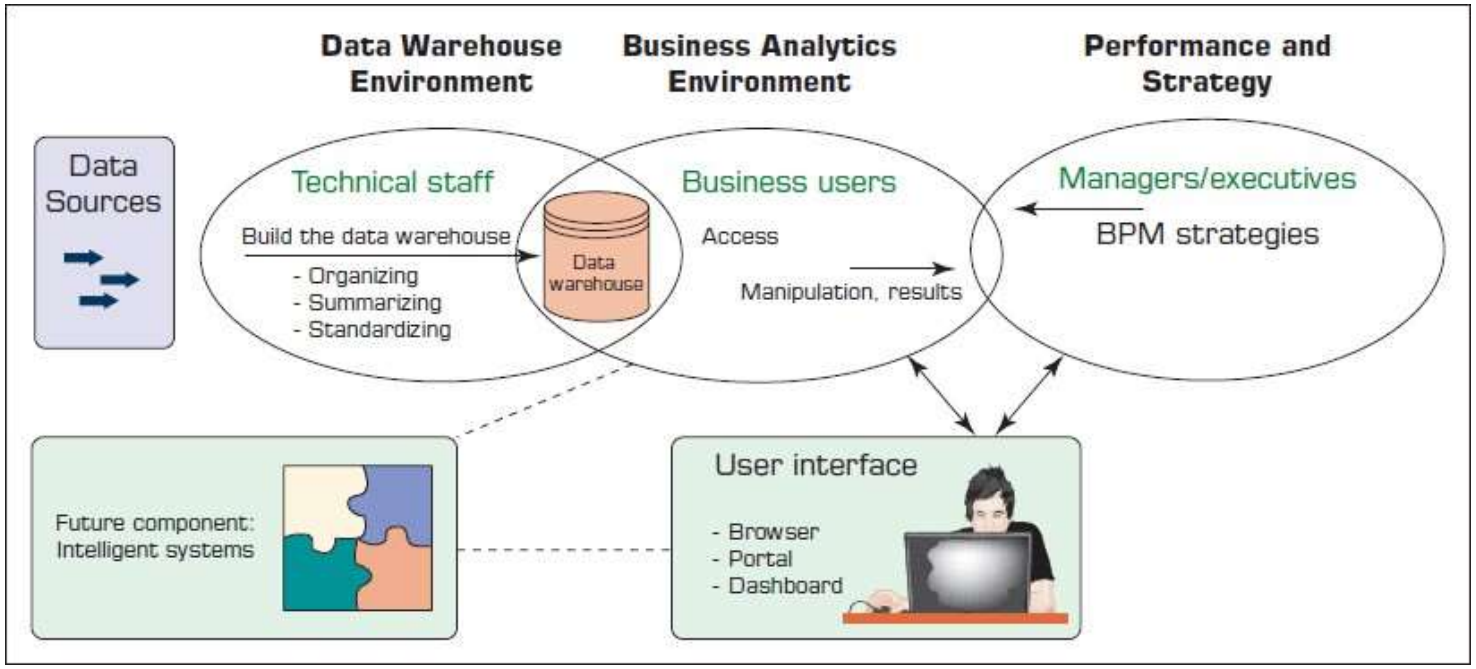


FIGURE 1.6 Evolution of Business Intelligence (BI).

Architecture of BI

BI has four major components:

1. **DW** (with its source data)
2. **Business analytics** (a collection of tools for manipulating, mining, and analyzing the data in the DW)
3. **BPM** (for monitoring & analyzing performance)
4. **User interface**



The Origins and Drivers of BI

- Business cycle times are extremely compressed; faster, and more informed.
- Managers need the right information at the right time and in the right place (core of the modern approaches to BI).
- Organizations are being driven to capture, understand, and harness their data to support decision making and improve business operations.
- Legislation/regulations require leaders to document their business processes and sign off on the validity of the information they report to stakeholders.

Data Warehouse (DW) as a Foundation for BI

- BI systems rely on DWs as the information source for creating insight and supporting managerial decisions.
- **DW** is a subject-oriented, integrated, time-variant, nonvolatile collection of data in support of management's decision-making process.

The three main types of data warehouses are:

- **Data Marts (DM):** a subset of a DW, typically consisting of a single subject area
- **Operational Data Stores (ODS):** provides a form of customer information file whose contents are updated throughout the course of business operations.
- **Enterprise Data Warehouses (EDW):** a large-scale data warehouse used across the enterprise for decision support.

Data Warehouse (DW) as a Foundation for BI (cont.)

- Data from many different sources can be extracted, transformed, and loaded into a DW for further access and analytics for decision support.

Data is structured to be available in a form ready for analytical processing activities:

- OLAP,
- data mining,
- querying,
- reporting,
- decision support applications

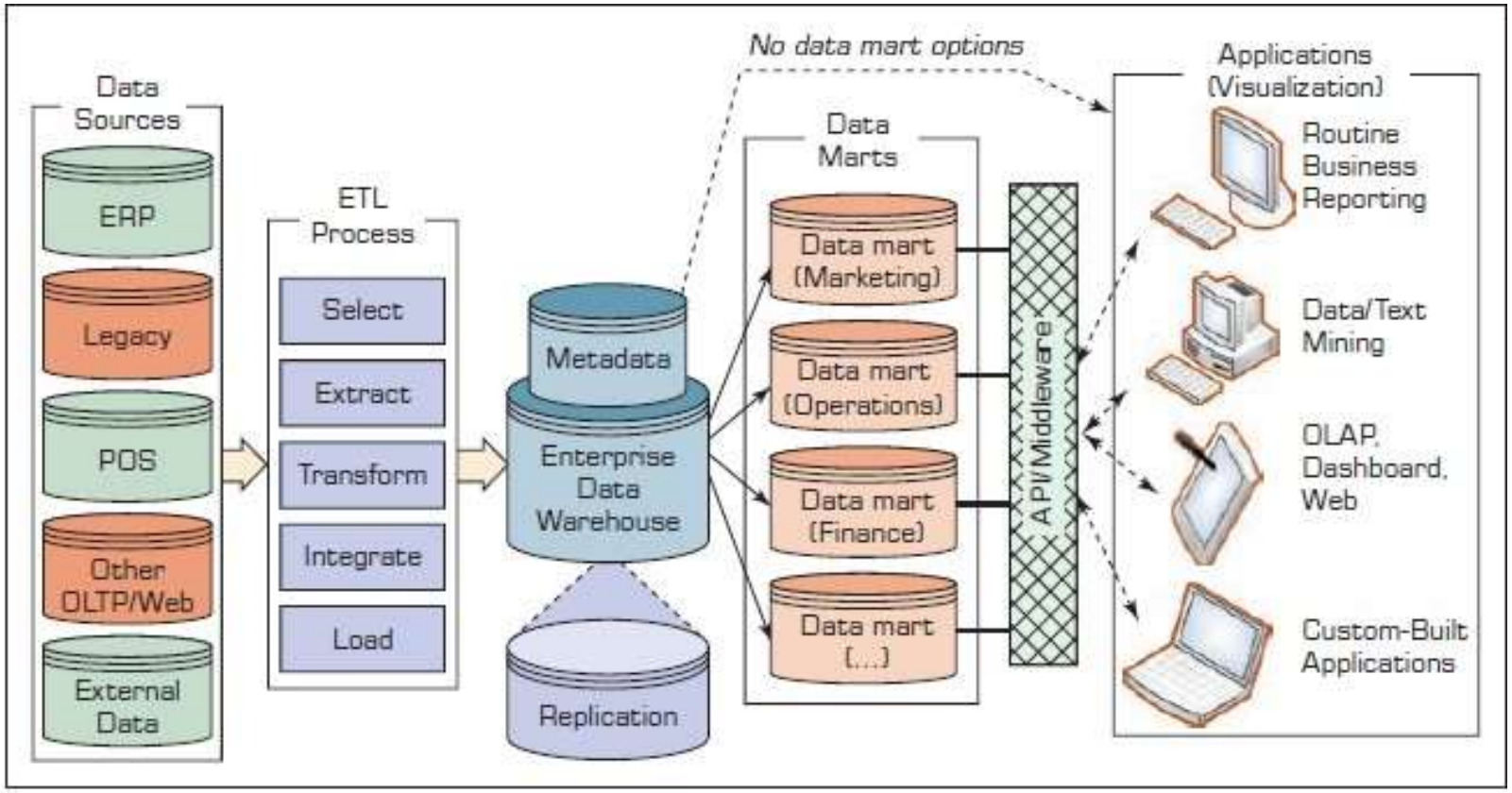


FIGURE 1.8 Data Warehouse Framework and Views.

Transaction Processing versus Analytic Processing

- **Transaction processing systems** are constantly involved in handling updates to what may be known as operational databases.
- **OLTP** (OnLine Transaction Processing) systems handle a company's routine ongoing business, where the computer responds immediately to user requests.
- The OLTP system is efficient for transaction processing, but inefficient for end-user ad hoc reports, queries, and analysis.
- In contrast, a **DW** is typically a distinct system that provides storage for data that will be used for analysis by **OLAP** (OnLine Analytical Processing) systems.

Multimedia Exercise in Business Intelligence

- The **fundamental reasons** for investing in BI must be aligned with the company's business strategy.
- **BI** improve company business processes and transforming it to be more data driven.

BI tools are sometimes integrated among themselves, resulting in six key trends:

1. Big Data
2. Focus on customer experience as opposed to just operational efficiency.
3. Mobile and even newer user interfaces—visual, voice, mobile.
4. Predictive and prescriptive analytics, machine learning, artificial intelligence.
5. Migration to cloud.
6. Much greater focus on security and privacy protection.

1.5 Analytics Overview

- Overview
- Descriptive, Predictive, Prescriptive Analytics
- Big Data

Analytics Overview

- **Analytics** is the process of developing actionable/recommendations for actions based on insights generated from historical data.

To solve real problems analytics represents the combination of

  - computer technology +
  - management science techniques +
  - statistics.

Descriptive, Predictive, Prescriptive Analytics

Three types of analytics:

1. **Descriptive**
2. **Predictive**
3. **Prescriptive**

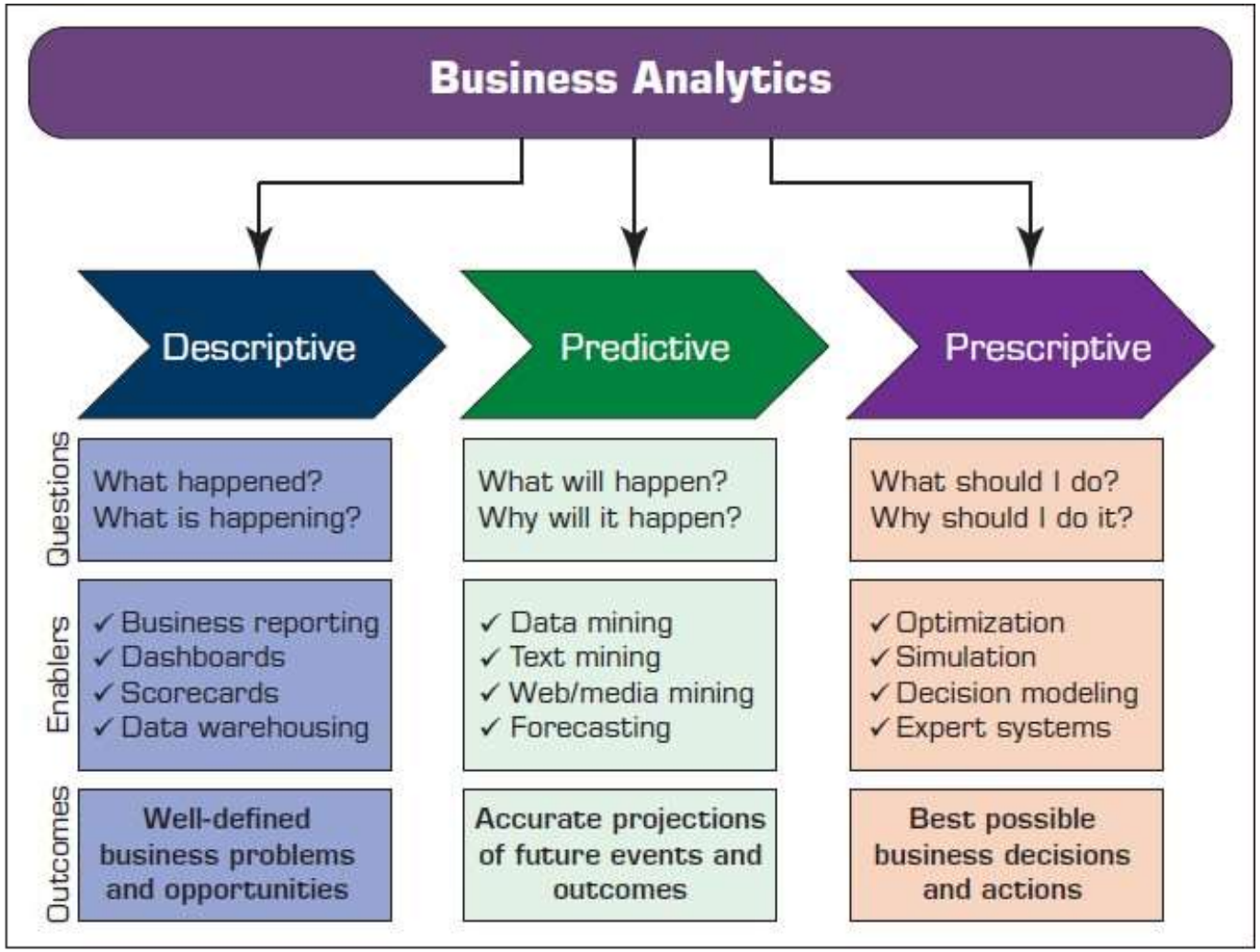


FIGURE 1.9 Three Types of Analytics.

Big Data

- **Big Data** refers to data that cannot be stored in a single storage unit.
- Different **forms**:
  - structured
  - unstructured
- **Characteristics** of big data:
  - volume
  - velocity
  - variety
- These are evolving quickly to encompass stream analytics, IoT, cloud computing, and deep learning– enabled AI.

Big Data (cont.)

There are two aspects to managing data on this scale:

1. **Storing:**
   - an extremely expensive storage solution **vs.** store data in chunks on different machines connected by a network.
   - Hadoop Distributed File System (HDFS): store a copy or two of this chunk in different locations on the network, both logically and physically.
2. **Processing:**
   - To process vast amounts of data **computation done by one powerful computer vs.** processing data sets with a parallel distributed algorithm on a cluster
   - MapReduce programming paradigm

1.6 Analytics Examples in Selected Domains

- Sports Analytics
- Sports Analytics Example 1: The Business Office
- Sports Analytics Example 2: The Coach
- Healthcare Analytics Example 1: Humana
- Retail Analytics
- Image Analytics

Sports Analytics

- **Sport analytics** focuses on gathering data about athletes and teams to create insights that improve sports decisions, such as
  - deciding which players to recruit,
  - how much to pay them,
  - who to play, etc.

- For teams, it involves business decisions such as
  - ticket pricing
  - roster /schedule decisions,
  - analysis of each competitor's strengths and weaknesses

- Sport is a big business.

Sports Analytics Example 1: The Business Office

- Dave works as a business analyst for a major pro baseball team, focusing on revenue.
- His responsibility include analyzing ticket sales; why season ticket holders renew (or do not renew) their tickets as well as what factors drive last-minute individual seat ticket purchases, and ticket pricing.
- Some of the analytical techniques Dave uses include simple statistics on behavior such as overall attendance and answers to survey questions about likelihood to purchase again.
- Dave uses dynamic pricing. This is a rich research area for many sports teams and has huge upside potential for revenue enhancement. For example, his pricing takes into account the team's record, who they are playing, game dates and times, etc.
- Dave builds regression models to pick out key factors driving these historic behaviors and create PMs (Predictive Models) to identify how to spend marketing resources to drive revenues. He builds churn models for season ticket holders to create segments of customers who will renew, will not renew, or are fence-sitters, which then drives more refined marketing campaigns.

Sports Analytics Example 2: The Coach

- Bob Breedlove is the football coach for a major college team. His areas of focus include recruiting the best high school players, developing them to fit his offense and defense systems, and getting maximum effort from them on game days. Sample questions in his area of responsibility include: Whom do we recruit? What drills help develop their skills?

- Dar Beranek specializes in helping the coaches make tactical decisions. She is working with a team of student interns who are creating opponent analytics. They used the coach's annotated game film to build a cascaded decision tree model to predict whether the next play will be a running play or passing play.

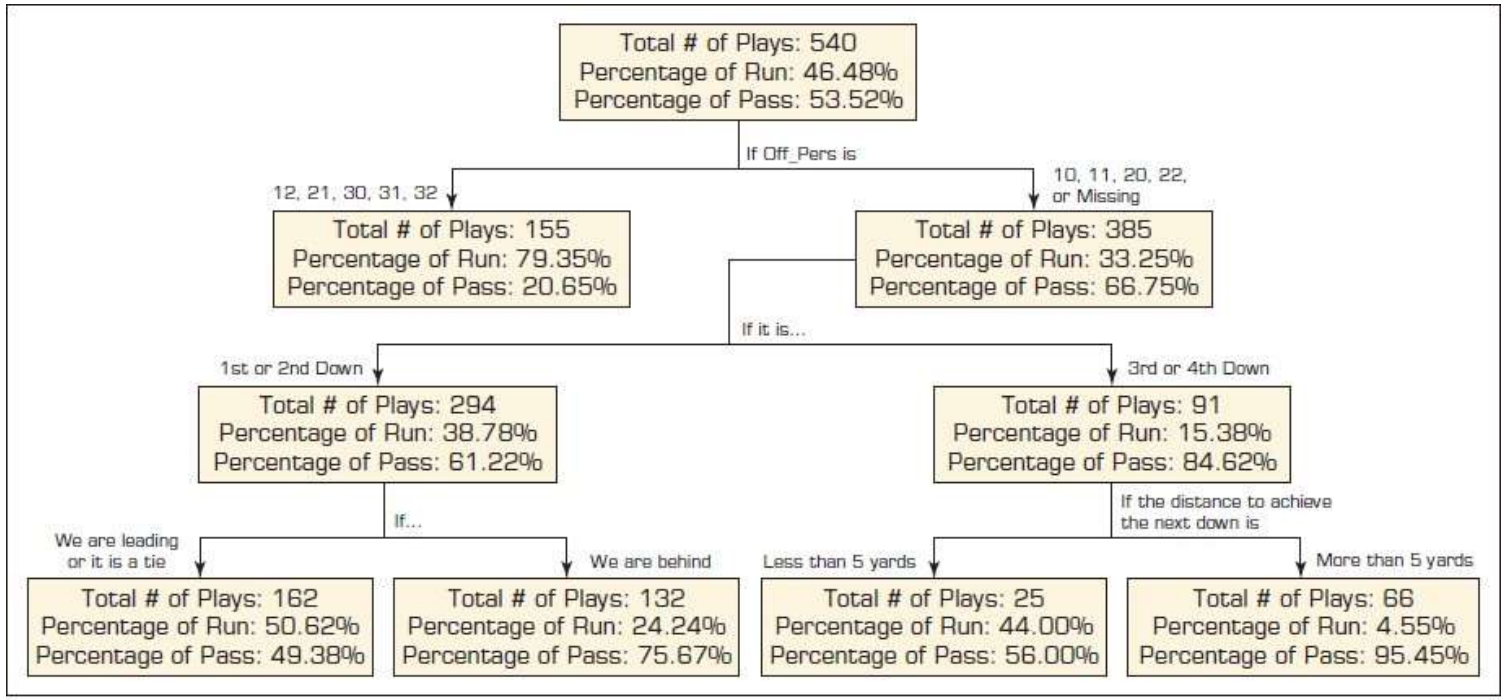


FIGURE 1.12 Cascaded Decision Tree for Run or Pass Plays.

Healthcare Analytics Example 1: Preventing Falls in Seniors

- Accidental falls are a major health risk for adults age 65+ years. The costs of falls pose a significant strain on the healthcare system; estimated to be $34 billion in 2013 alone.
- Humana is a major health insurance company, with 3.2 million members, most of whom are seniors. Keeping its senior members well and helping them live safely is a key business objective. Fall prevention is an important component.
- As such, there is a need for an accurate method to identify individuals at greatest risk of falling so that they can be proactively managed for fall prevention. Administrative medical and pharmacy claims, clinical data, temporal clinical patterns, and other data is needed to identify individuals at high risk of falling over a time horizon.
- Today, the Falls PM (Predictive Models) is central to Humana's ability to identify seniors who could benefit from fall mitigation interventions. Using the PM, Humana was able to identify 20,000 consumers at a high risk of falling who benefited from this program. Identified consumers wear a device that detects falls and alerts a service for immediate assistance.

Retail Analytics

- Goal of the analytics industry is to use information about business suppliers, customers, employees, and stakeholders to make better decisions.

- Example:
  - Amazon's enormous investments in analytics to power their value chain.
  - Walmart & Target have invested millions of dollars in analytics for their supply chains.

- Analytics—descriptive, predictive, or prescriptive—can play a role in making better data-driven decisions.

Retail Analytics (cont.)

Examples of key questions that can be answered through analytics, and the potential business value derived from fielding such analytics.

TABLE 1.1 Examples of Analytics Applications in the Retail Value Chain

Analytic Application	Business Question	Business Value
Inventory Optimization	1. Which products have high demand? 2. Which products are slow moving or becoming obsolete?	1. Forecast the consumption of fast-moving products and order them with sufficient inventory to avoid a stock out scenario. 2. Perform fast inventory turnover of slow-moving products by combining them with one in high demand.
Price Elasticity	1. How much net margin do I have on the product? 2. How much discount can I give on this product?	1. Markdown prices for each product can be optimized to reduce the margin dollar loss. 2. Optimized price for the bundle of products is identified to save the margin dollar.
Market-Basket Analysis	1. What products should I combine to create a bundle offer? 2. Should I combine products based on slow-moving and fast-moving characteristics? 3. Should I create a bundle from the same category or a different category line?	1. The affinity analysis identifies the hidden correlations between the products, which can help in following values: a. Strategize the product bundle offering based on focus on inventory or margin. b. Increase cross-selling or up-selling by creating bundle from different categories or the same categories, respectively.
Shopper Insight	1. Which customer is buying what product at what location?	1. By customer segmentation, the business owner can create personalized offers resulting in better customer experience and retention of the customer.
Customer Churn Analysis	1. Who are the customers who will not return? 2. How much business will I lose? 3. How can I retain the customers? 4. What demography of customer is my loyal customer?	1. Businesses can identify the customer and product relationships that are not working and show high churn. Thus, they can have better focus on product quality and the reason for that churn. 2. Based on the customer lifetime value (LTV), the business can do targeted marketing resulting in retention of the customer.

Retail Analytics (cont.)

TABLE 1.1 Examples of Analytics Applications in the Retail Value Chain

Analytic Application	Business Question	Business Value
Channel Analysis	1. Which channel has lower customer acquisition cost? 2. Which channel has better customer retention? 3. Which channel is more profitable?	1. Marketing budget can be optimized based on insight for better return on investment.
New Store Analysis	1. What location should I open? 2. What and how much opening inventory should I keep?	1. Best practices of other locations and channels can be used to get a jump-start. 2. Comparison with competitor data can help to create a differentiator to attract the new customers.
Store Layout	1. How should I do store layout for better topline? 2. How can I increase my in-store customer experience?	1. Understand the association of products to decide store layout and better alignment with customer needs. 2. Workforce deployment can be planned for better customer interactivity and thus satisfying customer experience.
Video Analytics	1. What demography is entering the store during the peak period of sales? 2. How can I identify a customer with high LTV at the store entrance so that a better personalized experience can be provided to this customer?	1. In-store promotions and events can be planned based on the demography of incoming traffic. 2. Targeted customer engagement and instant discount enhances the customer experience resulting in higher retention.

Image Analytics

- The benefits of satellite data are significant to scientists who need to regularly monitor global change, land usage, and weather.

Examples of image analytics include:

  - **World Bank** researchers used satellite data to propose strategic recommendations for urban planners and officials from developing nations.
  - **EarthCast** provides accurate weather updates for a large commercial U.S. airline. It map out conditions along a flight path.
  - **Imazon** started using satellite data to develop a picture of close real-time information on Amazon deforestation.

Reference

- Ramesh Sharda, Dursun Delen, Efraim Turban; Analytics, Data Science, & Artificial Intelligence: Systems For Decision Support, 11ed, Global Edition, Pearson Education Limited, 2021, ISBN 10: 1 - 2 92-34155-6, ISBN 13: 978-1-292-34155-2

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Recommended Video

- Understanding Decision Support Systems

  https://www.youtube.com/watch?v=k31CSoO7wC4

- What is Business Intelligence (BI)?

  https://www.youtube.com/watch?v=hDJdkcdG1iA